NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

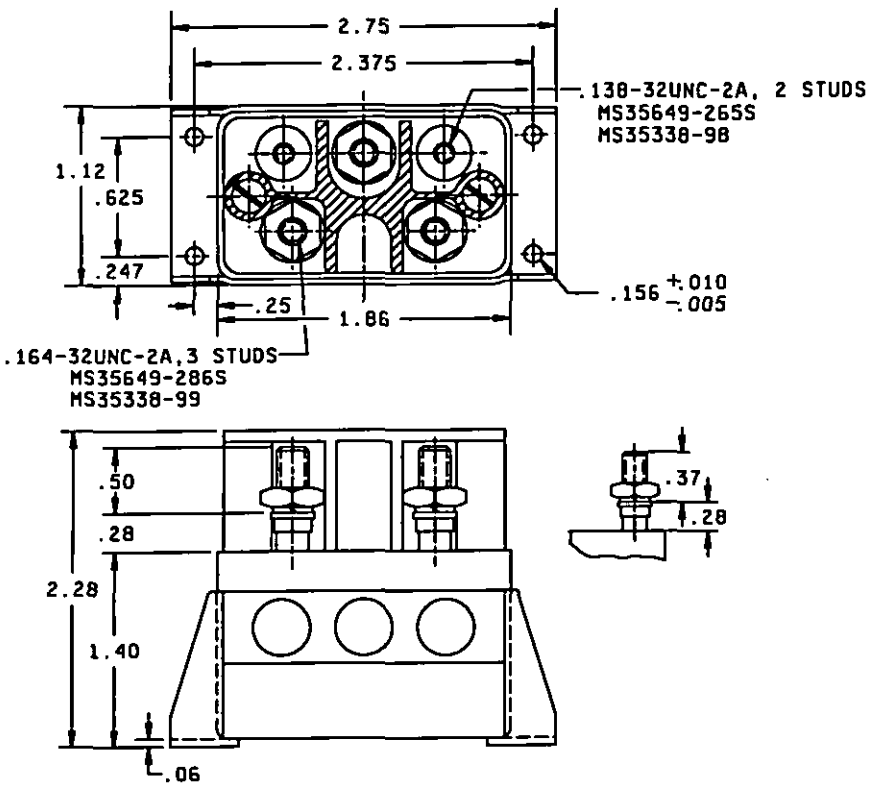
MIL-PRF-6106/15C(USAF)
11 February 1986
SUPERSEDING
MIL-R-6106/15B(USAF)
14 August 1984

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, 50 AMPERES, DC, SPDT, HERMETICALLY SEALED,

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

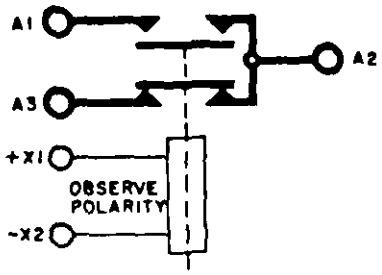


Inches	mm
.005	0.13
.010	0.25
.06	1.52
.13	3.30
.138	3.51
.156	3.96
.164	4.17
.247	6.27
.25	6.35
.28	7.11
.37	9.40
.50	12.70
.625	15.88
1.12	28.45
1.40	35.56
1.86	47.24
2.28	50.91
2.375	60.33
2.75	69.85
3.96	100.58

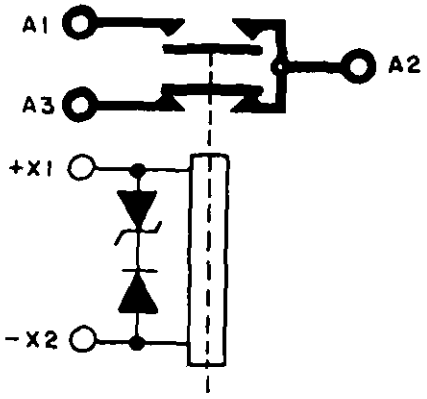
FIGURE 1. Relay, outline drawing (-001 and -002).

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

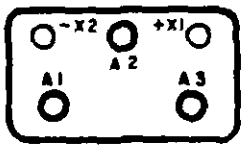
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CIRCUIT DIAGRAM-001



CIRCUIT DIAGRAM-002
(SEE NOTE 6)

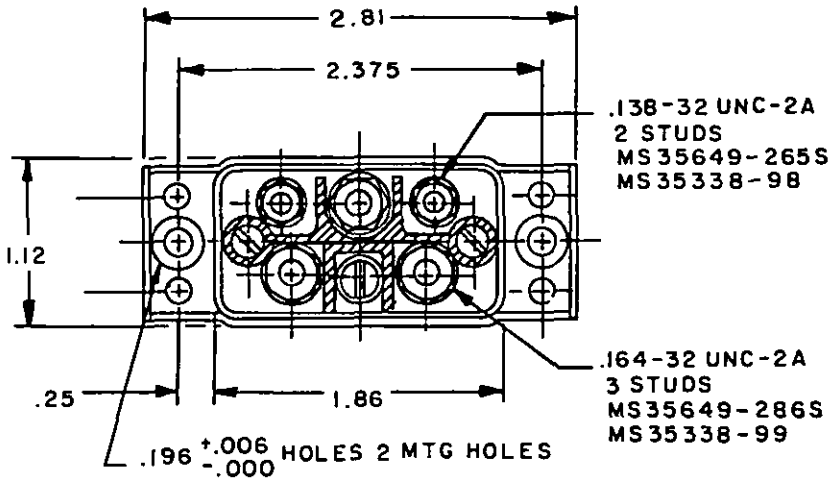


TERMINAL LAYOUT

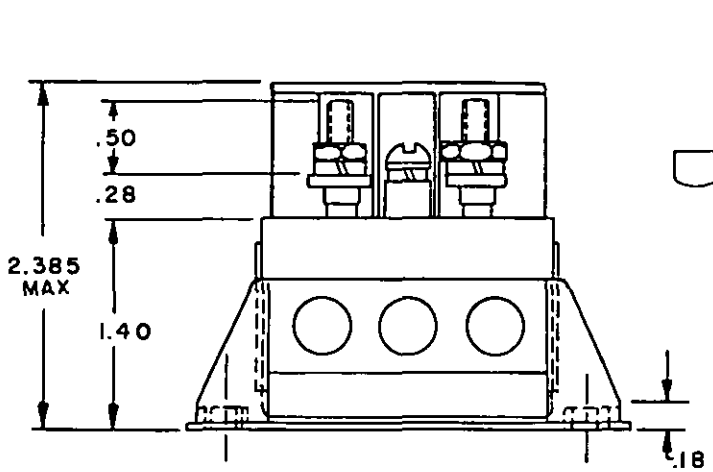
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.03 (.76 mm) for two place decimals and ±.010 (.25 mm) for three place decimals.
4. Terminal identification and locations designated by diagram on relay case.
5. Relay shall not operate or be damaged by reverse coil polarity.
6. Coil transient voltage limited to ±42 volts referenced to zero volts.

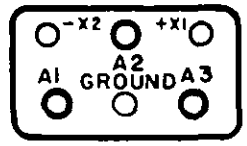
FIGURE 1. Relay, outline drawing (-001 and -002) - Continued.

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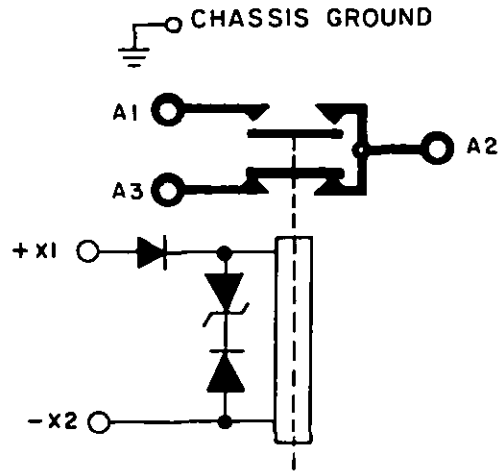


Inches	mm
.006	0.15
.138	3.51
.164	4.17
.18	4.6
.196	4.98
.25	6.4
.28	7.11
.37	9.4
.50	12.7
1.12	28.5
1.40	35.6
1.86	47.2
2.375	60.33
2.385	60.58
2.81	71.4





TERMINAL LAYOUT



CIRCUIT DIAGRAM-003
(SEE NOTE 6)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.03 (.76 mm) for two place decimals and ±.010 (.25 mm) for three place decimals.
4. Terminal identification and locations designated by diagram on relay case.
5. Relay shall not operate or be damaged by reverse coil polarity.
6. Coil transient voltage limited to ±42 volts referenced to zero volts.

© FIGURE 2. Relay outline drawing (-003).

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REQUIREMENTS:

Weight: 0.38 pound (172.4 grams) maximum.

Temperature range: -70°C to +125°C.

Maximum altitude rating: 80,000 feet.

Shock G-level: 25 G.

Duration: 11 ms.

Maximum duration contact opening: 10 µs.

Vibration-sinusoidal:

G-level: 15 G.

Frequency range: 10 to 2,000 Hz.

Acceleration: 15 G.

Insulation resistance: 100 megohms.

Dielectric withstanding voltage (sea level):

All terminals (initial and after life tests): 1,050 V rms.

Dielectric withstanding voltage (altitude):

All terminals (initial and after life tests): 500 V rms.

Maximum contact drop initial: 0.150 volt.

After life test: 0.175 volt.

Overload current: 400 amperes dc.

Rupture current: 500 amperes dc.

Duty rating: Continuous.

Part number: M6106/15 (and a dash number from table I).

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© TABLE I. Dash number and operating characteristics.

Dash no.	Coil data								Time-milliseconds-max 5/		
	Nominal voltage (V dc) 1/	Max		Max pick-up voltage			Drop-out voltage 3/ 4/	Hold voltage 3/ 4/	Operate	Release	Bounce
		V dc	Amp 2/	Normal 3/ 4/	High temp. test	Cont. current test					
-001	28 V dc	30	.114	18	20	20	1.5	7.0	25	25	2
-002	28 V dc	30	.114	18	20	20	1.5	7.0	25	25	2
-003	28 V dc	32	.120	18	20	20	1.5	7.0	50	40	2

1/ CAUTION: Use of any coil voltage less than nominal voltage will compromise the operation of the relay.
2/ At 25°C.
3/ Pickup, hold, and dropout voltages shown are for test purposes only and shall not be used for design criteria.
4/ Over the temperature range.
5/ With nominal coil voltage.

TABLE II. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles x 10^3	28 V dc			
		Main		Aux	
		NO	NC	NO	NC
Resistive	50	50	50	---	---
Inductive	50	50	50	---	---
Motor	50	20	20	---	---
Lamp	50	20	20	---	---
Mechanical life, reduced current	100	12.5	12.5	---	---
Minimum current	50	5	5	---	---

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